

Month	Week	Topic	Subtopics	Tools/Libraries	Outcome
Month 1	Week 1	Python Basics	Variables, Data Types, Operators, Type Conversion, Input Output, Comments	Python	Write simple Python programs
	Week 1	Control Flow	If statements, Nested conditions, Boolean logic	Python	Write decision-based programs
	Week 2	Loops	For loop, While loop, Break, Continue	Python	Iterate through data
	Week 2	Functions	User-defined functions, Parameters, Return values, Lambda functions	Python	Write reusable functions
	Week 3	Data Structures	Lists, Tuples, Dictionaries, Sets, List comprehension	Python	Efficient data handling
	Week 3	File Handling	Read files, Write files, CSV files	Python	Process datasets
	Week 4	Object Oriented Programming	Classes, Objects, Constructors, Inheritance, Polymorphism	Python	Write structured code
	Week 4	NumPy Basics	Arrays, Indexing, Slicing, Broadcasting, Vectorization	NumPy	Numerical computations
	Week 4	Pandas Basics	Series, DataFrames, Data loading, Filtering, Groupby	Pandas	Data analysis skills
	Week 4	Data Visualization	Line plots, Histograms, Scatter plots, Heatmaps	Matplotlib, Seaborn	Visualize data insights
Month 2	Week 5	ML Introduction	Supervised vs Unsupervised learning, ML pipeline	Scikit-learn	Understand ML workflow
	Week 5	Regression Algorithms	Linear Regression, Polynomial Regression, Ridge, Lasso	Scikit-learn	Build regression models
	Week 6	Classification Algorithms	Logistic Regression, KNN, Naive Bayes, SVM	Scikit-learn	Build classification models
	Week 6	Tree Models	Decision Trees, Random Forest	Scikit-learn	Understand tree models
	Week 7	Clustering	K-Means, Hierarchical Clustering, DBSCAN	Scikit-learn	Unsupervised learning
	Week 7	Dimensionality Reduction	PCA, Feature selection	Scikit-learn	Improve model performance
	Week 8	Model Evaluation	Confusion Matrix, Precision, Recall, F1 Score, ROC-AUC	Scikit-learn	Evaluate ML models
	Week 8	Deep Learning Basics	Neural Networks, Activation functions, Loss functions	TensorFlow / PyTorch	Understand neural networks
Month 3	Week 9	NLP Basics	Text preprocessing, Tokenization, Stopwords removal	NLTK	Process text data
	Week 9	Text Processing	Stemming, Lemmatization, N-grams	NLTK, spaCy	Clean text data
	Week 10	Vectorization	Bag of Words, TF-IDF	Scikit-learn	Convert text to numbers
	Week 10	Word Embeddings	Word2Vec, GloVe	Gensim	Understand semantic meaning
	Week 11	NLP Applications	Sentiment analysis, Text classification	Python NLP libraries	Build NLP models
	Week 11	Named Entity Recognition	Entity detection	spaCy	Information extraction
	Week 12	Generative AI Basics	Prompt engineering, Text generation	OpenAI API	Understand GenAI systems
	Week 12	Embeddings & Vector Search	Semantic search, embeddings	Hugging Face	Build GenAI apps
Month 4	Week 13	Transformer Architecture	Self-attention, Multi-head attention, Positional encoding	PyTorch, Transformers	Understand transformers
	Week 14	Large Language Models	BERT, GPT, LLaMA	Hugging Face	Work with LLMs
	Week 14	Fine Tuning LLMs	Prompt tuning, Parameter efficient tuning	Hugging Face	Customize LLMs
	Week 15	RAG Systems	Vector embeddings, semantic search	LangChain, LlamaIndex	Build RAG pipelines
	Week 15	Vector Databases	Similarity search, indexing	FAISS, Pinecone	Store embeddings
	Week 16	Agentic AI	Autonomous agents, tools, memory	CrewAI, LangGraph	Build AI agents
	Week 16	Latest AI Tools	Multimodal AI, AI automation	LangChain ecosystem	Understand modern AI stack
Month 5	Week 17	ML Project	End-to-end ML project pipeline	Python, Scikit-learn	Portfolio project
	Week 18	NLP Project	Sentiment analysis or text classifier	Python NLP libraries	NLP project
	Week 19	LLM Project	PDF chatbot using RAG	LangChain, Vector DB	GenAI project
	Week 20	Computer Vision Project	Image classification CNN	TensorFlow / PyTorch	CV project

	Week 20	Model Deployment	Web app deployment	Streamlit	Deploy ML app
Month 6	Week 21	Interview Preparation	ML theory questions	ML concepts	Interview readiness
	Week 22	Coding Practice	Data structures and coding problems(Optional)	LeetCode, HackerRank	Improve coding
	Week 23	Portfolio Building	GitHub projects, documentation	GitHub	Professional portfolio
	Week 24	Job Applications	Resume preparation, LinkedIn networking	LinkedIn, Job portals	Apply for AI/ML roles